

VONEGAN, Russian Federation

WMO: 238230

Lat: 61.9561N

Lon: 64.9378E

Elev: 111

StdP: 99.99

Time Zone: 5.00 (E05)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -39.0	(c) -35.4	(d) -41.7	(e) 0.1	(f) -38.8	(g) -38.2	(h) 0.1	(i) -35.1	(j) 8.0	(k) -10.4	(l) 7.2	(m) -13.7	(n) 1.3	(o) 300	(p) 0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.6	(c) 28.3	(d) 18.7	(e) 26.4	(f) 17.9	(g) 24.5	(h) 17.1	(i) 20.0	(j) 26.1	(k) 18.9	(l) 24.8	(m) 17.9	(n) 23.1	(o) 3.0	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	12.9	22.7	16.7	12.0	21.4	15.7	11.3	20.7	57.9	25.7	54.3	24.8	50.9	22.9	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.5	(b) 6.4	(c) 5.6	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 7.5	(b) 6.4	(c) 5.6	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.9	-20.3	-14.9	-7.1	-0.4	6.8	15.5	16.9	13.1	7.0	-1.4	-11.0	-16.3		
	DBStd	13.97	8.78	8.04	6.83	5.40	5.40	4.66	4.38	4.00	3.74	5.39	7.95	8.97		
	HDD10.0	4543	938	696	529	313	134	8	3	16	104	354	631	814		
	HDD18.3	7100	1197	929	788	563	361	110	82	167	339	612	881	1073		
	CDD10.0	552	0	0	0	1	34	173	216	113	15	1	0	0		
	CDD18.3	68	0	0	0	0	2	24	36	6	0	0	0	0		
	CDH23.3	661	0	0	0	0	38	242	327	54	0	0	0	0		
	CDH26.7	143	0	0	0	0	5	46	83	9	0	0	0	0		
	WSAvg	2.4	2.1	2.3	2.7	2.9	2.7	2.5	2.2	2.0	2.5	2.8	2.4	2.2		
	PrecAvg	661	40	35	43	43	52	72	72	74	60	64	58	46		
(15) Precipitation	PrecMax	756	66	64	90	81	96	130	120	123	99	112	90	76		
	PrecMin	517	16	7	6	1	22	39	26	40	26	26	10	27		
	PrecStd	55	14	13	20	17	18	25	28	21	23	22	17	13		
	PrecStd	55	14	13	20	17	18	25	28	21	23	22	17	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.2	1.2	8.3	14.9	27.4	30.5	31.1	28.0	19.8	14.6	4.3	0.1		
		MCWB	-2.2	-0.3	2.2	9.3	16.1	20.1	20.3	19.3	13.0	10.3	3.4	-0.8		
	2%	DB	-3.4	-0.4	5.3	11.4	24.1	28.0	28.8	24.6	17.3	10.0	1.1	-1.2		
		MCWB	-4.1	-1.7	1.2	6.0	14.1	18.6	19.0	17.7	11.8	6.9	0.0	-1.9		
	5%	DB	-6.0	-2.2	3.6	9.5	20.6	26.1	26.9	22.2	15.1	7.9	0.0	-3.2		
		MCWB	-6.7	-3.3	0.2	4.8	12.5	17.4	18.6	15.8	10.9	5.6	-1.0	-3.9		
	10%	DB	-8.4	-4.2	1.9	7.6	17.3	24.2	24.8	20.3	13.2	5.7	-1.3	-5.3		
		MCWB	-9.0	-5.1	-0.7	3.7	10.8	16.4	17.9	15.0	10.0	4.3	-2.1	-5.8		
	0.4%	WB	-2.1	0.2	2.7	10.2	17.1	21.4	21.2	20.0	13.9	11.1	3.5	-0.7		
		MCDB	-1.3	1.1	6.9	15.0	26.1	27.1	28.3	26.3	17.8	14.4	4.5	-0.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-4.0	-1.5	1.6	7.0	14.9	19.9	20.3	18.2	12.5	7.5	0.4	-1.8		
		MCDB	-3.4	-0.5	4.8	10.4	22.7	25.7	26.6	23.2	16.1	9.6	1.0	-1.1		
	5%	WB	-6.5	-3.3	0.5	5.2	13.1	18.4	19.3	17.0	11.2	5.7	-0.9	-3.8		
		MCDB	-5.9	-2.2	3.1	9.0	19.2	24.9	25.1	20.7	14.8	7.5	-0.1	-3.3		
	10%	WB	-8.9	-5.2	-0.5	3.7	11.3	17.3	18.5	15.8	10.0	4.4	-2.1	-5.8		
		MCDB	-8.3	-4.2	1.9	7.0	17.1	22.9	23.7	19.3	12.3	5.7	-1.3	-5.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.4	9.7	9.2	10.3	11.0	9.6	9.3	7.5	5.6	6.8	7.0		
		MCDBR	8.9	8.1	9.3	11.2	15.1	13.6	13.5	12.9	10.5	7.5	5.9	6.0		
		MCWBR	8.6	7.3	6.3	7.1	7.7	6.1	5.6	6.5	6.2	5.2	5.7	5.6		
		MCDBR	9.2	7.9	8.6	9.8	13.8	12.1	12.1	10.9	9.4	6.9	6.0	6.0		
	5% WB	MCWBR	8.9	7.1	6.2	6.6	7.5	6.0	5.6	6.2	6.0	5.0	5.9	5.7		
		MCWBR	8.9	7.1	6.2	6.6	7.5	6.0	5.6	6.2	6.0	5.0	5.9	5.7		
(40) Clear-Sky Solar Irradiance	taub		0.242	0.292	0.337	0.371	0.375	0.389	0.419	0.399	0.350	0.313	0.256	0.230		
	taud		1.973	2.041	2.017	2.090	2.334	2.367	2.276	2.310	2.417	2.334	2.055	1.915		
	Ebn at Noon		415	639	743	804	842	836	793	776	755	638	396	234		
	Edh at Noon		39	76	112	128	110	109	116	104	77	57	35	23		
(44) All-Sky Solar Radiation	RadAvg		0.32	1.18	2.70	4.16	4.91	5.30	5.22	3.68	2.18	1.01	0.34	0.15		
	RadStd		0.02	0.09	0.16	0.53	0.41	0.37	0.47	0.35	0.29	0.13	0.05	0.01		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page